Emergency Signal Guide

How to signal for help if you become lost or injured in the backcountry of Pierce County

If you become lost or injured in the backcountry, how you signal for help can dramatically affect how quickly rescuers find you. As volunteers who search across 1,800 square miles of Pierce County — from dense forest to alpine glaciers — we know that effective signaling cuts rescue times significantly. SAR teams report that 80% of lost hikers are found within 1 mile of their last known position when they stay put.

> **FIRST PRIORITY**
> If you have ANY cell service, call 911. Give: your location (GPS coordinates if possible), the nature of your emergency, number of people in your group, injuries, and what you're wearing. Even with weak signal, try texting 911 — texts require far less signal than calls. Turn on location services on your phone.

S.T.O.P. — What to Do When Lost

S — Sit Down

Stop moving immediately. Panic is your biggest enemy. The moment you realize you're lost, stop walking. Every step in the wrong direction takes you farther from where rescuers will look for you. Take a deep breath. Drink some water. Eat a snack.

T — Think

When did you last know where you were? Can you identify any landmarks — a ridge, a stream, a trail junction? What time is it? How much daylight do you have? What resources do you have with you?

O — Observe

Look around carefully. Check your map and compass. Can you hear a road, a river, or other hikers? Can you see a trail marker, a power line, or a familiar peak? Look for your own footprints to retrace.

P — Plan

If you can safely and confidently retrace your steps to the last known point, do so. If there is ANY doubt, stay where you are. Make yourself visible and comfortable. Begin signaling. If you filed a trip plan, rescuers will start looking along your expected route. Be there when they arrive.

> **SAR PRO TIP**
> The single most important thing you can do when lost: STAY PUT. Moving makes it exponentially harder for search teams to find you. We search your planned route first. If you've wandered miles from it, you've made our job much harder.

Audio Signals

Whistle (Best Option)

Three short blasts, pause, repeat. A whistle at 100+ decibels carries up to a mile — far farther than your voice — and requires almost no energy. This is why a whistle is one of the most important items in your pack. Plastic whistles work when wet and in freezing temperatures (don't use metal — it can freeze to your lips).

Voice

If you don't have a whistle, shout 'HELP' in groups of three. Cup your hands around your mouth and aim toward sounds of searchers.

Conserve your voice — shouting is exhausting and can cause you to lose it entirely. A whistle is far superior.

Visual Signals

Signal Mirror

A signal mirror flash can be seen up to 10 miles away on a clear day and is one of the most effective long-range signaling tools. Aim by holding up a peace sign, placing the target (aircraft, ridge) between your fingers, and tilting the mirror until the reflected light hits your fingers. Sweep slowly back and forth. Any shiny surface works: a phone screen, foil, polished metal, even a credit card.

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Bright Colors & Movement

Wear or display brightly colored clothing or gear in an open area. Spread out a bright tent, tarp, or sleeping bag. Movement catches the eye — wave a bright item overhead when you hear or see searchers. Arrange gear in patterns visible from the air.

Fire & Smoke

Fire is visible at night; smoke during the day. Build signal fires in open areas on high ground if safe. For daytime visibility, add green branches or damp material for white smoke (visible against dark backgrounds). Three fires in a triangle is an international distress signal. Only build fires where safe — never in dry, windy conditions.

Ground-to-Air Symbols

Create large symbols visible from aircraft using rocks, branches, or by stamping in snow. Make them at least 10 feet tall with high contrast to the ground. Key symbols:

- X = Need medical assistance
- V = Need help / assistance
- -> = Traveling this direction
- SOS = Universal distress signal

Each line should be at least 3 feet wide (5:1 length-to-width ratio minimum).

Electronic Signaling

Cell Phone

Even with no signal bars, try calling 911 — emergency calls can connect through ANY available carrier, not just yours. If calling fails, try texting 911 (available in most areas). Move to higher ground or an opening for better signal. Conserve battery: switch to airplane mode between attempts, reduce screen brightness, close apps.

Personal Locator Beacon (PLB)

A PLB sends your GPS location via satellite to global rescue coordination centers. Works anywhere on earth with no subscription. When activated, STAY IN PLACE — rescuers are heading to the exact coordinates your beacon transmitted. Cost: $250-350, no ongoing fees. The best investment a backcountry traveler can make.

Satellite Communicator

Devices like Garmin inReach or SPOT allow two-way text messaging and SOS via satellite anywhere in the world. These require a monthly subscription ($12-65/mo) but let you communicate your specific situation and receive instructions from rescuers. Two-way communication can dramatically improve rescue efficiency.

KEY TAKEAWAYS

1) Call 911 if you have any signal.
2) STAY PUT — don't wander.
3) Make yourself visible: bright colors, open ground, signals.
4) Three of anything is a distress signal.
5) A $5 whistle is your best friend.
6) If you filed a trip plan, we're already looking for you.